

1) The Load Impedance (Z_L) in ohm, that absorbs the maximum average power is :

Prob. 1: Consider the circuit shown in Fig. 1.

1. The load impedance (Z_L), in Ω , that absorbs the maximum average power is:

2. The open circuit voltage between a-b terminals (V_s), when the load impedance is disconnected, is:

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3. The short circuit current (I_{sc}), when a-b terminals are shorted is:

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4. The maximum average power transferred to the load impedance (Z_L) is:

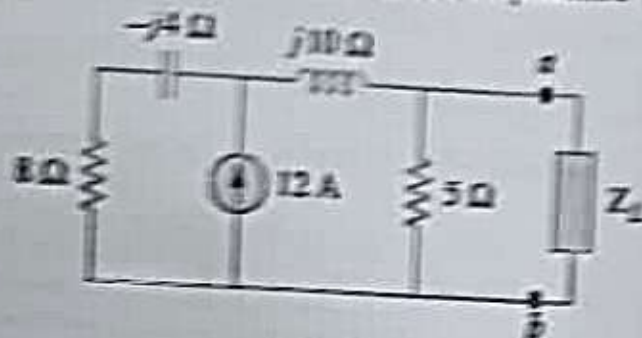


Fig.1

4) The maximum average power transferred to the load impedance (Z_L) is: (8 marks)

Prob. 2: Consider the circuit shown in Fig. 2 and using mesh analysis method,

1. Write the two mesh equations, do not simplify:

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2. The current I_1 is: 2) The current i_1 :

3. The current I_2 is: 3) The current i_2 :

4. The capacitor current, I_c , is: 4) The current i_c :

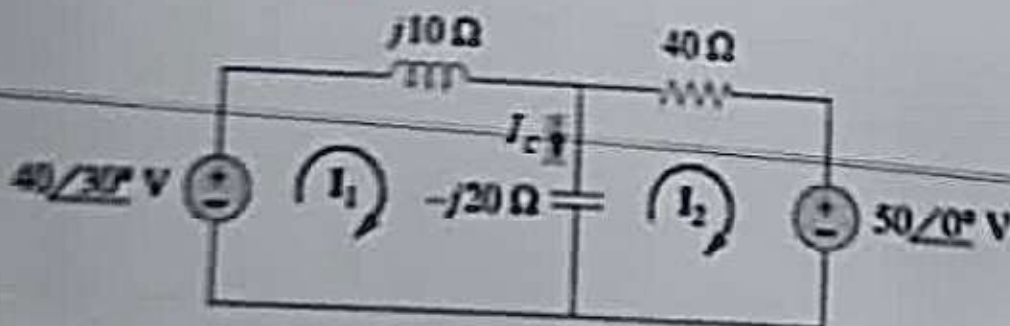


Fig 2

(8 marks)

Prob 3: Two loads are connected in parallel across a voltage source of $100 \angle 0^\circ V_{rms}$, where load 1 absorbs 500VAR at 0.6 lagging p.f. while load 2 absorbs 400VA at 0.75 leading p.f

1. The impedance of load 1 (Z_1), in Ω , is:
2. The current of load 2 (I_2), in Amps, is:
3. The complex power (S) supplied by the source is:
4. The power factor of the equivalent load is:
5. The capacitance needed to raise the pf to unity is:
6. The current supplied by the source, after the power factor is corrected, is:

(12 marks)

Prob. 4: Consider the circuit shown in Fig. 3,

1. The equivalent impedance seen by the source, Z_{in} , is:
2. The current supplied by the source, I_1 , is:
3. The load current, I_2 , is:
4. The voltage across the primary winding, V_1 , is:
5. The average power absorbed by the 2Ω resistor is:

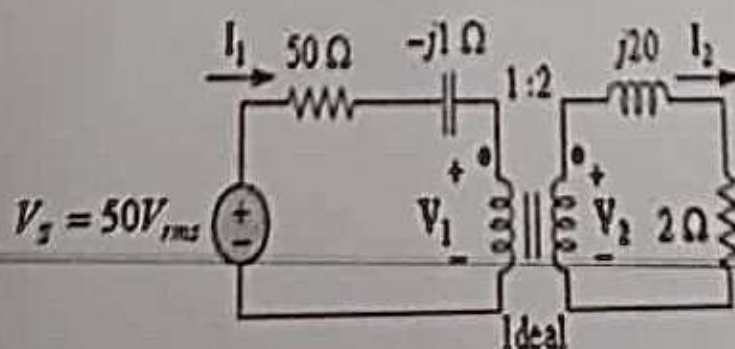
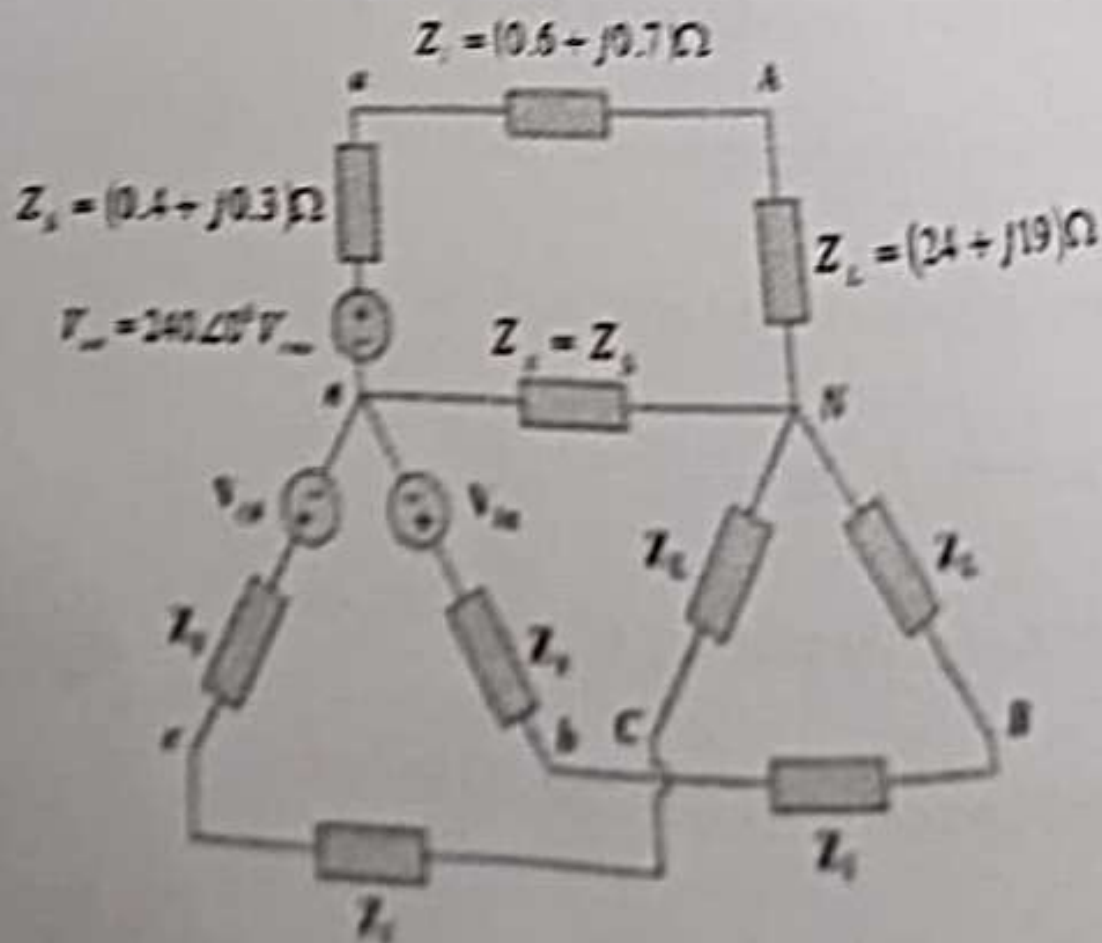


Fig.3

(10 marks)

Prob. 5: Fig. 4 shows a balanced, positive sequence, Y-connected voltage source with $V_{an} = 240 \angle 0^\circ V_{rms}$ and internal impedance of $Z_s = (0.4 + j0.3) \Omega$ per phase, connected to a Y-connected balanced load with an impedance of $Z_L = (24 + j19) \Omega$ per phase. If the line joining the source and the load has an impedance of $Z_l = (0.6 + j0.7) \Omega$ per phase,

1. The line current I_{aA} is:
2. The load phase voltage V_{AN} is:
3. The load line voltage V_{AB} is:
4. The source line voltage V_{ab} is:
5. The complex power supplied by phase a of the source is:
6. The total average power absorbed by the load is:



* Prop 1^o

$$(1) Z_L = 3.41 - 0.73j$$

$$(2) V_o = 37.5 \angle -51.34^\circ$$

$$(3) I_{sc} = 10.73 \angle -63.43^\circ$$

$$(4) P_{max} = 51.55 \text{ watt}$$

* Prop 2^o

$$(1) (10j - 20j)I_1 + 20jI_2 = 40 \angle 30^\circ$$

$$20jI_1 + (40 - 20j)I_2 = 50 \angle 180^\circ$$

$$(2) I_1 = 4.69 \angle 95.2^\circ$$

$$(3) I_2 = 0.99 \angle 37.69^\circ$$

$$(4) I_c = 4.24 \angle 106.56^\circ$$

* Prop 3^o

$$(1) Z_1 = 9.6 + 12.8j$$

$$(2) I_2 = 6.047 \angle 41.41^\circ$$

$$(3) P = 834.5 \angle 6.88^\circ$$

$$(4) P_f = 0.9886 \text{ Lagging}$$

$$(5) C = 40.17 \text{ } \mu\text{F}$$

$$(6) I = 8.285 \angle 0^\circ$$

* Prop 4^o

$$(1) Z_{in} = 50.5 + 4j$$

$$(2) I_1 = 0.987 \angle -4.53^\circ$$

$$(3) I_2 = 0.4935 \angle -4.53^\circ$$

$$(4) V_1 = 4.96 \angle 79.76^\circ$$

$$(5) P = 0.487 \text{ watt}$$

Laith Q Amro

Prop 580

$$\textcircled{1} \quad I_{aA} = 7.496 \angle -38.66^\circ$$

$$\textcircled{2} \quad V_{AN} = 229.5 \angle -0.292^\circ$$

$$\textcircled{3} \quad V_{AB} = 397.5 \angle 29.708^\circ$$

$$\textcircled{4} \quad V_{ab} = 409.2 \angle 30.03^\circ$$

$$\textcircled{5} \quad S_p = \overset{1771}{\cancel{1771}} \angle 38.66^\circ$$

$$\textcircled{6} \quad P = 4046.4 \text{ watt}$$

Laith Q Amro ↵